

Clustering Review Questions and Answers

Based on EECS 836 Course Material by Zijun Yao

May 2025

Introduction

This document provides a comprehensive set of fundamental, computational, and conceptual questions and answers based on the “10-Clustering.pdf” lecture slides from the EECS 836: Machine Learning course at the University of Kansas, taught by Assistant Professor Zijun Yao. The questions cover clustering concepts, K-means, K-means++, Gaussian Mixture Models (GMM), and the Expectation-Maximization (EM) algorithm, as presented in the document.

1 Clustering Concepts

1.1 Fundamental Questions

1. **Question:** What is the primary goal of clustering?
Answer: Clustering aims to group a set of objects such that objects in the same cluster are similar to one another and different from objects in other clusters (page 3).
2. **Question:** How does clustering relate to unsupervised learning?
Answer: Clustering is a common form of unsupervised learning, where patterns are learned from raw, unlabeled data without given classifications, unlike supervised learning (page 4).
3. **Question:** What are two main types of clustering algorithms mentioned in the document?
Answer: The two main types are partitional clustering, which divides data into non-overlapping subsets, and hierarchical clustering, which organizes clusters into a nested tree structure (page 10).
4. **Question:** What is a well-separated cluster?
Answer: A well-separated cluster is a set of points where each point is closer to every other point in the same cluster than to any point outside it (page 12).
5. **Question:** What defines a prototype-based cluster?
Answer: A prototype-based cluster is a set of objects closer to the cluster’s prototype (centroid or medoid) than to the prototypes of other clusters (page 13).

6. **Question:** What is a contiguity-based cluster?
Answer: A contiguity-based cluster is a set of points where each point is closer to at least one point in the same cluster than to any point outside it (page 14).
7. **Question:** What characterizes a density-based cluster?
Answer: A density-based cluster is a dense region of points separated by low-density regions, useful for irregular shapes and handling noise (page 15).
8. **Question:** What is the difference between exclusive and non-exclusive clustering?
Answer: Exclusive clustering assigns points to only one cluster, while non-exclusive clustering allows points to belong to multiple clusters (page 11).
9. **Question:** What is fuzzy clustering?
Answer: Fuzzy clustering assigns each point to every cluster with a weight or probability between 0 and 1, with weights summing to 1 (page 11).
10. **Question:** Why is defining similarity important in clustering?
Answer: Defining similarity (e.g., distance or proximity between vectors) is crucial because it determines how objects are grouped based on their relatedness, depending on the data and task (page 9).

1.2 Computational Questions

1. **Question:** Given points (1, 2), (3, 4), and (5, 6) in a cluster, compute the centroid.
Answer: The centroid is the mean of the points (page 13). For (1, 2), (3, 4), (5, 6):
 $x\text{-coordinate} = \frac{1+3+5}{3} = 3,$
 $y\text{-coordinate} = \frac{2+4+6}{3} = 4.$
 Thus, the centroid is (3, 4).
2. **Question:** For a point $x = (2, 3)$ and centroids $c_1 = (1, 1)$, $c_2 = (4, 5)$, compute the Euclidean distance to each centroid.
Answer: Euclidean distance is used in K-means (page 20). For $x = (2, 3)$:
 To $c_1 = (1, 1)$: $\sqrt{(2-1)^2 + (3-1)^2} = \sqrt{1+4} = \sqrt{5} \approx 2.236.$
 To $c_2 = (4, 5)$: $\sqrt{(2-4)^2 + (3-5)^2} = \sqrt{4+4} = \sqrt{8} \approx 2.828.$
3. **Question:** In K-means++ (page 30), if the first centroid is chosen and the squared distances from points x_1, x_2, x_3 to it are 9, 16, and 25, what is the probability of selecting x_2 as the next centroid?
Answer: The probability is $\frac{\min d^2(C_j, x_i)}{\sum_i \min d^2(C_j, x_i)}$ (page 30). Squared distances: 9, 16, 25.
 Sum = $9 + 16 + 25 = 50$. For x_2 : $\frac{16}{50} = 0.32$.
4. **Question:** For a cluster with points (2, 2), (4, 4), compute the Sum of Squared Error (SSE) if the centroid is (3, 3).
Answer: SSE is the sum of squared distances to the centroid (page 20). For centroid (3, 3):
 Distance to (2, 2): $\sqrt{(3-2)^2 + (3-2)^2} = \sqrt{2}$, squared = 2.
 Distance to (4, 4): $\sqrt{(3-4)^2 + (3-4)^2} = \sqrt{2}$, squared = 2.
 SSE = $2 + 2 = 4$.

5. **Question:** If a K-means run has 100 points, 5 clusters, 10 iterations, and 2 attributes, estimate the computational complexity.
Answer: Complexity is $O(n \cdot K \cdot I \cdot d)$ (page 19). For $n = 100$, $K = 5$, $I = 10$, $d = 2$:
 $100 \cdot 5 \cdot 10 \cdot 2 = 10,000$ operations.
6. **Question:** In a GMM with $K = 2$, if mixing coefficients are $\pi_1 = 0.4$, $\pi_2 = 0.6$, and a point has probabilities $p(x | z = 1) = 0.5$, $p(x | z = 2) = 0.3$, compute $p(x)$.
Answer: $p(x) = \sum_{k=1}^K \pi_k p(x | z = k)$ (page 39).
 $p(x) = 0.4 \cdot 0.5 + 0.6 \cdot 0.3 = 0.2 + 0.18 = 0.38$.
7. **Question:** In EM, if responsibilities for a point are $r_1^{(i)} = 0.7$, $r_2^{(i)} = 0.3$, and the point is $x^{(i)} = (1, 2)$, compute the contribution to the new mean μ_1 .
Answer: The mean update is $\mu_k = \frac{\sum_{i=1}^N r_k^{(i)} \cdot x^{(i)}}{\sum_{i=1}^N r_k^{(i)}}$ (page 48). Contribution to μ_1 :
 $r_1^{(i)} \cdot x^{(i)} = 0.7 \cdot (1, 2) = (0.7, 1.4)$.
8. **Question:** If a GMM has $\pi_1 = 0.5$, and responsibilities $r_1^{(1)} = 0.8$, $r_1^{(2)} = 0.2$, compute the updated π_1 .
Answer: $\pi_k = \frac{1}{N} \sum_{i=1}^N r_k^{(i)}$ (page 48). For $N = 2$:
 $\pi_1 = \frac{0.8+0.2}{2} = \frac{1}{2} = 0.5$.
9. **Question:** For points $(0, 0)$, $(1, 1)$, assign them to the closest centroid from $c_1 = (0, 1)$, $c_2 = (2, 2)$ using Euclidean distance.
Answer: Compute distances (page 16):
For $(0, 0)$: To c_1 : $\sqrt{(0-0)^2 + (0-1)^2} = 1$; To c_2 : $\sqrt{(0-2)^2 + (0-2)^2} = \sqrt{8} \approx 2.828$. Assign to c_1 .
For $(1, 1)$: To c_1 : $\sqrt{(1-0)^2 + (1-1)^2} = 1$; To c_2 : $\sqrt{(1-2)^2 + (1-2)^2} = \sqrt{2} \approx 1.414$. Assign to c_1 .
10. **Question:** In K-means, if a cluster has points $(1, 1)$, $(2, 2)$, and the centroid moves to $(1.5, 1.5)$, compute the change in centroid coordinates.
Answer: Old centroid: $(\frac{1+2}{2}, \frac{1+2}{2}) = (1.5, 1.5)$. New centroid: $(1.5, 1.5)$. Change: $(1.5 - 1.5, 1.5 - 1.5) = (0, 0)$. (Note: This assumes the centroid is already at the mean; if not, specify prior centroid.)

1.3 Conceptual Questions

1. **Question:** Why is clustering considered subjective?
Answer: Clustering is subjective because the notion of a cluster can be ambiguous, and different groupings (e.g., based on gender or occupation) may be valid depending on the context or criteria, as shown with the Simpson's family example (page 8).
2. **Question:** How does the choice of similarity measure impact clustering results?
Answer: The similarity measure (e.g., Euclidean distance, cosine similarity) defines how relatedness is quantified, affecting cluster assignments. Choosing an inappropriate measure for the data or task can lead to poor or meaningless clusters (page 9).

3. **Question:** Why is the number of clusters K a critical parameter in K-means?
Answer: The number of clusters K must be specified in K-means and directly affects the granularity of the clustering. Too few or too many clusters can lead to under- or over-segmentation, missing natural groupings or creating trivial clusters (page 16).
4. **Question:** Why is K-means sensitive to initial centroid selection?
Answer: K-means converges to a local minimum of the Sum of Squared Error (SSE), and different initial centroids can lead to different local minima, resulting in varied cluster assignments, as shown in the examples of suboptimal clustering (pages 22–24).
5. **Question:** How does the K-means++ algorithm improve upon standard K-means initialization?
Answer: K-means++ selects initial centroids to be far apart by choosing the first randomly and subsequent ones with probability proportional to their squared distance from existing centroids, reducing the chance of poor clustering and improving SSE (page 30).
6. **Question:** Why does K-means struggle with non-globular cluster shapes?
Answer: K-means assumes clusters are spherical and centered around centroids, using Euclidean distance. Non-globular shapes (e.g., elongated or irregular clusters) lead to poor assignments, as points may be closer to a wrong centroid (page 34).
7. **Question:** How can K-means limitations be addressed for differing cluster sizes?
Answer: To handle differing cluster sizes, one can use a large number of small clusters to capture parts of natural clusters, followed by post-processing to merge them into larger, natural clusters (page 32).
8. **Question:** Why is the EM algorithm suitable for Gaussian Mixture Models?
Answer: The EM algorithm is suitable for GMMs because it handles latent variables (cluster assignments) by iteratively computing posterior probabilities (E-step) and updating model parameters (M-step) to maximize the likelihood, addressing the chicken-and-egg problem of unknown assignments and parameters (pages 46–47).
9. **Question:** How does the probabilistic nature of GMMs differ from the deterministic assignments in K-means?
Answer: GMMs assign points to clusters probabilistically using mixing coefficients and Gaussian distributions, allowing soft assignments, while K-means assigns points deterministically to the nearest centroid, resulting in hard assignments (pages 39, 49).
10. **Question:** Why is marginalizing out latent variables in GMMs computationally challenging?
Answer: Marginalizing out latent variables (cluster assignments) in GMMs involves summing over all possible assignments, placing the log outside the sum in the likelihood function, which prevents simplification and makes direct optimization difficult, necessitating the EM algorithm (page 46).

2 K-means and K-means++

2.1 Fundamental Questions

1. **Question:** What is the main objective of the K-means algorithm?
Answer: The K-means algorithm aims to minimize the sum of distances (typically squared Euclidean distances) from each point to its assigned cluster centroid (page 16).
2. **Question:** What is a centroid in K-means?
Answer: A centroid is the mean of all points assigned to a cluster, representing the cluster's center (page 19).
3. **Question:** What is the role of the initial centroids in K-means?
Answer: Initial centroids determine the starting points for cluster assignments, and the algorithm iteratively updates them to form clusters (page 16).
4. **Question:** What is the Sum of Squared Error (SSE) in K-means?
Answer: SSE is the sum of squared distances from each point to its nearest centroid, used as the objective function to minimize in K-means (page 20).
5. **Question:** How does K-means typically stop iterating?
Answer: K-means stops when the centroids no longer change or when relatively few points change clusters (page 19).

2.2 Computational Questions

1. **Question:** Given centroids (2, 2), (6, 6), and points (3, 3), (5, 5), assign each point to a cluster.
Answer: Use Euclidean distance (page 16):
For (3, 3): To (2, 2): $\sqrt{(3-2)^2 + (3-2)^2} = \sqrt{2} \approx 1.414$; To (6, 6): $\sqrt{(3-6)^2 + (3-6)^2} = \sqrt{18} \approx 4.243$. Assign to (2, 2).
For (5, 5): To (2, 2): $\sqrt{(5-2)^2 + (5-2)^2} = \sqrt{18} \approx 4.243$; To (6, 6): $\sqrt{(5-6)^2 + (5-6)^2} = \sqrt{2} \approx 1.414$. Assign to (6, 6).
2. **Question:** If a cluster has points (1, 0), (0, 1), (2, 1), compute the new centroid after one iteration.
Answer: Centroid is the mean (page 16):
 $x\text{-coordinate} = \frac{1+0+2}{3} = 1$,
 $y\text{-coordinate} = \frac{0+1+1}{3} = \frac{2}{3} \approx 0.667$.
New centroid: (1, 0.667).
3. **Question:** In K-means++, if squared distances to the first centroid are 4, 9, 16 for three points, compute the probability of selecting the third point as the next centroid.
Answer: Probability: $\frac{\min d^2(C_j, x_i)}{\sum_i \min d^2(C_j, x_i)}$ (page 30). Sum = 4 + 9 + 16 = 29. For third point: $\frac{16}{29} \approx 0.5517$.
4. **Question:** Compute the SSE for a cluster with points (0, 0), (2, 2), and centroid (1, 1).
Answer: SSE = sum of squared distances (page 20):

To (0, 0): $\sqrt{(1-0)^2 + (1-0)^2} = \sqrt{2}$, squared = 2.

To (2, 2): $\sqrt{(1-2)^2 + (1-2)^2} = \sqrt{2}$, squared = 2.

SSE = 2 + 2 = 4.

5. **Question:** If K-means runs for 8 iterations with 50 points, 3 clusters, and 4 attributes, estimate the complexity.

Answer: Complexity: $O(n \cdot K \cdot I \cdot d)$ (page 19).

$50 \cdot 3 \cdot 8 \cdot 4 = 4,800$ operations.

2.3 Conceptual Questions

1. **Question:** Why does K-means converge quickly in the first few iterations?

Answer: Most of the reduction in SSE occurs early as points are reassigned to closer centroids, and subsequent iterations make smaller adjustments (page 19).

2. **Question:** How does the choice of K affect the interpretability of K-means results?

Answer: Choosing K determines the number of clusters, affecting whether the results align with meaningful patterns in the data. Incorrect K can lead to uninterpretable or trivial clusters (page 21).

3. **Question:** Why do outliers pose a problem for K-means?

Answer: Outliers can distort centroids, as they contribute to the mean calculation, pulling the centroid away from the true cluster center and leading to suboptimal clustering (page 31).

4. **Question:** How does K-means++ avoid choosing outliers as initial centroids?

Answer: K-means++ selects centroids with probability proportional to their squared distance from existing centroids, favoring points far from others but avoiding extreme outliers due to the probabilistic selection (page 30).

5. **Question:** Why might multiple K-means runs be necessary?

Answer: Due to sensitivity to initial centroids, different runs may yield different results. Multiple runs allow selection of the clustering with the lowest SSE, improving result quality (page 29).

3 Gaussian Mixture Models and EM Algorithm

3.1 Fundamental Questions

1. **Question:** What is a Gaussian Mixture Model (GMM)?

Answer: A GMM represents a data distribution as a weighted sum of K Gaussian distributions, each with mixing coefficient π_k , mean μ_k , and covariance Σ_k (page 39).

2. **Question:** What is a latent variable in the context of GMMs?

Answer: A latent variable, such as the cluster assignment z , is an unobserved variable indicating which Gaussian component generated a data point (page 38).

3. **Question:** What are mixing coefficients in a GMM?
Answer: Mixing coefficients π_k represent the probability of choosing the k -th Gaussian component, summing to 1 and non-negative (page 39).
4. **Question:** What is the purpose of the Expectation-Maximization (EM) algorithm?
Answer: The EM algorithm iteratively estimates parameters of models with latent variables, like GMMs, by alternating between computing posterior probabilities (E-step) and updating parameters (M-step) (page 47).
5. **Question:** What does the E-step compute in the EM algorithm for GMMs?
Answer: The E-step computes the posterior probability (responsibility) $r_k^{(i)}$ that component k generated data point i , given current parameters (page 48).

3.2 Computational Questions

1. **Question:** In a GMM with $K = 3$, if $\pi_1 = 0.3$, $\pi_2 = 0.5$, $\pi_3 = 0.2$, and conditional probabilities are $p(x | z = 1) = 0.4$, $p(x | z = 2) = 0.6$, $p(x | z = 3) = 0.1$, compute $p(x)$.
Answer: $p(x) = \sum_{k=1}^K \pi_k p(x | z = k)$ (page 39).
 $p(x) = 0.3 \cdot 0.4 + 0.5 \cdot 0.6 + 0.2 \cdot 0.1 = 0.12 + 0.3 + 0.02 = 0.44$.
2. **Question:** If responsibilities for two points are $r_1^{(1)} = 0.6$, $r_1^{(2)} = 0.4$, compute the updated mixing coefficient π_1 for $K = 2$.
Answer: $\pi_k = \frac{1}{N} \sum_{i=1}^N r_k^{(i)}$ (page 48). For $N = 2$:
 $\pi_1 = \frac{0.6+0.4}{2} = 0.5$.
3. **Question:** For a point $x^{(i)} = (2, 3)$ with responsibilities $r_1^{(i)} = 0.8$, $r_2^{(i)} = 0.2$, compute its contribution to the covariance matrix Σ_1 if $\mu_1 = (1, 1)$.
Answer: Contribution to Σ_1 : $r_1^{(i)}(x^{(i)} - \mu_1)(x^{(i)} - \mu_1)^T$ (page 48).
 $x^{(i)} - \mu_1 = (2 - 1, 3 - 1) = (1, 2)$.
 $(x^{(i)} - \mu_1)(x^{(i)} - \mu_1)^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.
Contribution: $0.8 \cdot \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 0.8 & 1.6 \\ 1.6 & 3.2 \end{bmatrix}$.
4. **Question:** If a GMM has $N = 3$ points with responsibilities $r_1^{(1)} = 0.5$, $r_1^{(2)} = 0.3$, $r_1^{(3)} = 0.2$, compute the denominator for μ_1 update.
Answer: Denominator: $\sum_{i=1}^N r_k^{(i)}$ (page 48).
 $0.5 + 0.3 + 0.2 = 1.0$.
5. **Question:** If a point has posterior probabilities $p(z = 1 | x) = 0.7$, $p(z = 2 | x) = 0.3$, what are the responsibilities?
Answer: Responsibilities $r_k^{(i)} = p(z^{(i)} = k | x; \theta)$ (page 48).
 $r_1^{(i)} = 0.7$, $r_2^{(i)} = 0.3$.

3.3 Conceptual Questions

1. **Question:** Why does the EM algorithm alternate between E-step and M-step?
Answer: Alternating between E-step (inferring latent variables) and M-step (up-

dating parameters) addresses the difficulty of optimizing both simultaneously, iteratively improving the likelihood by leveraging current estimates (page 47).

2. **Question:** How does the generative process of GMMs differ from K-means?

Answer: GMMs model data as generated from a mixture of Gaussians with probabilistic assignments, while K-means assumes deterministic assignments to centroids, lacking a probabilistic generative model (pages 40, 49).

3. **Question:** Why is maximum likelihood used to fit GMM parameters?

Answer: Maximum likelihood chooses parameters that maximize the probability of observing the data, ensuring the GMM closely matches the data's density (page 45).

4. **Question:** How do responsibilities in EM improve clustering over K-means?

Answer: Responsibilities provide soft assignments, allowing points to belong to multiple clusters with varying probabilities, capturing uncertainty and leading to more flexible clustering than K-means' hard assignments (page 48).

5. **Question:** Why is the log-likelihood in GMMs difficult to optimize directly?

Answer: The log-likelihood involves a sum inside the logarithm due to marginalizing latent variables, preventing simplification and requiring iterative methods like EM to approximate the solution (page 46).